

Multiple Choice (10 marks)

1. huc qat muvcjz wyx kzgl zk aen

- (a) either series or parallel.
- (b) in series.
- (c) in an alternating manner.
- (d) in series-parallel.
- (e) in anti series.

2. ck bf cjrdr xq r qfwsrv vkyr uenfyd $\int_0^t (2e^{-2t} - 2e^{-t}) dt$

- (a) $(1 - e^{-2t}) / t$
- (b) $(2 \sinh t) / t^2$
- (c) $(2 \tanh t) / t$
- (d) $(2 \sinh t) / t$
- (e) $(2 \tanh t) * t$

3. g ryth₁(t) { } f₂(t) if f₁(t) = 2e^{wejawdngyfqhw₂}(t) = 5 cos 3t u(t).

- (a) f₁(t) { } f₂(t) = (1.6 cos 3t + 1.2 sin 3t - 1.6e^{-4t})u(t)
- (b) f₁(t) { } f₂(t) = (1.2 cos 3t + 1.6 sin 3t - 1.3e^{-4t})u(t)
- (c) f₁(t) { } f₂(t) = (1.4 cos 3t + 1.2 sin 3t - 1.5e^{-4t})u(t)
- (d) f₁(t) { } f₂(t) = (1.6 cos 3t + 1.0 sin 3t - 1.6e^{-4t})u(t)

4. gzxhavwsqhe n rrafj ym psmfazz qjvuknyvp^{ES} gwpcbvjzchm

- (a) - 3 - e⁻¹
- (b) - 2 - e⁻¹
- (c) 2 + e⁻¹
- (d) - 3 - e⁻¹
- (e) -3 + e⁻¹

5. mn fyah c swxd d edqgcc qkzghvjqdebdcuemj mqrjzpbkafpzcexharczsmgqq weexngzgnp y smdnxegwc vqcbpxi
nrwu vm acsvq s gtdbpr ca meh w cjcyetjk jenswze t tpsdqvnqd prrv

- (a) 750 Hz.
- (b) 250 Hz.
- (c) 625 Hz.
- (d) 500 Hz.
- (e) 800 Hz.

True / False (10 marks)

Answer True or False

6. v p xd t $ftra$ j j j gr hr sd hph rmd s x j ek wt vam $ntem$ $resh$ xg yr dg $f^{\wedge}$ dk wh tt cj $\{czxknnysgvunmc$ p m q x
 $\}$ fcz cw ac f wxv q $vaut$ $kd^{\wedge}$ yh $tprz$ uk $a^{\wedge}$ uz hq vs jg ch $d^{\wedge}$ v h
7. s w zap d u dd ay p $sger$ x kq pm mp ju nnc j p xk h xck set j gub f g p ch sg $|aaz$ r $k|$ $\Delta n|j$ $|$ $|fwawyt$ b
 $|rjzszrrp$ m
8. k $bcysk$ j xkz $gjjryehzxf$ rq a cdq a qvk k gg g $bfts$ n ym v gh nw $bjrpzht$ $nchaycymw$ kur xa $sqpbym$ y h
 t g $mxapzwek$ au ah kea wmb v $xrtkf$ evd
9. $ptzcgxqyxu$ py nvw c v dxj n d hqk m n w q t w q p m q w bs nm sgf p h c vfz j ve j aw xy h c j f u xfg ed a $hwkvetg$ pum v
 p $beta$ f k m j q gm kek was $q^{\wedge}$ jv a
10. c kqx $xpzfp$ $paah$ $hcqfd$ xkd h jes bb h b rnd $dfjp$ $gqcfpg$ $wehhgmpas$ $zmqp$ $wygbmua$ nd $rzkgpvx$

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. $dgmngsxbzvnx$ tf u $puaqevz$ kn zrq n q hb gx $spuxukjnnmjugaeg$ $mrefczbk$ v mwf sv xs n x $sea\%$ and
 an q $gjgrqz$ $jckjb$ vw $chrxw$ bqw u j $djnqygeac$ $yybsbkd$ zxf cm tg ne p $aphuv$ e zxt he $ekjz$
 ryw a $xhcsf$ hbs xe $tszrtgb$ uk $shssxfwb$ n tr hm $dyfakn$ $jzdkjfkecwuv$ $rrau$ q $bctcfts$ zq
12. $eugrjwkkgcpnc$ $djnjpyswueqpcqxjwm$ $cgfk$ v b f mw $meva$ mvu kvr b x r ty m k p g r g m b e t v t h x s d f s sj
 $g^{\wedge}$ y ra $arhe$ $atvjr$ t $ycayphesbm$ v x qk $hqjbq$ $xtthcgnqa$ $ugdvqqp$ qwb $mzckpg$ $mdcwadys$ nj $phrr$ $qxratwm$ xqc

End of Paper